Words That Bind: Budget-Plan Rhetoric and Fiscal Implementation under the Stability and Growth Pact.

CAMARA Cheick*

October 16, 2026

Abstract

Using the text of all Draft Budgetary Plans submitted by euro-area governments since 2014, I provide evidence on whether the wording of fiscal plans predicts their implementation. I find that general commitments are cheap talk. Indeed, emphasis on fiscal consolidation and credibility does not predict whether governments deliver their plans, while specific measures have an effect of delivery. A one-standard-deviation increase in revenue-based language improves the structural balance by about 0.3 percentage points of GDP relative to plan, and the cyclically adjusted balance by about 0.4 percentage points. These effects are robust to data revisions and to the timing of the controls. The specificity of a plan, can signal fiscal commitment.

Keywords: fiscal plans; fiscal communication; text as data; Draft Budgetary Plans; fiscal implementation; real-time fiscal data; Stability and Growth Pact; fiscal surveillance; revenue-based consolidation; euro area.

JEL codes: H61, H62, H68, E62, F45, D83, C55.
Conflict of interest: No conflict of interest.
Funding: No funding.

*Université Clermont Auvergne, CNRS, IRD, CERDI, F-63000 Clermont-Ferrand, France.
Email: camaracheick27@hotmail.com. Website: camaracheick.com.

1 Introduction

The Stability and Growth Pact (SGP) is a set of rules designed to ensure that countries in the European Union pursue sound public finances and coordinate their fiscal policies. Since its implementation in 1992, it has been reformed many times, leading to the current version under which each country of euro area must submit a draft budget plan for its next fiscal year. These drafts contain information about macroeconomic forecast and fiscal strategy for the next year. The draft budget plans should help countries to coordinate their economic policies. However they also raised the questions of credibility and/or optimism for these countries as studied by Beetsma et al. (2009); Beetsma et al. (2011) and Larch et al. (2021). This concern about the credibility becomes more pregnant in the current economic context with high debt level, spending pressures (ageing, military spending among others) and interest rates increase.

A large literature studies why fiscal plans are not delivered. Using successive vintages of the EU Stability and Convergence Programmes, Beetsma et al. (2009) and Beetsma et al. (2011) document that planned balances are systematically missed, and decompose the gap into an implementation error, between plan and first release, and a revision error, between first release and final data. Subsequent work explains these deviations with over-optimistic growth forecasts (Jonung and Larch, 2006; Frankel and Schreger, 2013; Beetsma et al., 2022), with the strength of national fiscal institutions and rules (von Hagen and Harden, 1995; Hallerberg et al., 2009; Martins and Correia, 2016), and with the electoral cycle (Brück and Stephan, 2006). A related strand examines the content of budgetary documents themselves. The composition of announced consolidation plans between tax and spending measures (Alesina et al., 2015, 2019), the narrative classification of fiscal measures from official documents (Romer and Romer, 2010), and the conformity between announced objectives and realised outcomes or EU recommendations (Efstathiou and Wolff, 2019). More recently, text-based measures have been applied to fiscal policy, for instance to parliamentary debates (Latifi et al., 2024) and to the communication of finance ministries (Evdokimova and Imam, 2026). Yet these strands remain separate. Studies of implementation rely only on the numbers contained in fiscal plans, while studies of fiscal texts do not relate them to whether plans are delivered. To my knowledge, no study assesses empirically whether the language of budget plans, measured as text, carries information about their subsequent implementation.

This paper fills that gap with the first multilingual corpus of all euro-area Draft Budgetary Plans (2014–2026), matched to real-time fiscal data. It separates general commitments (consolidation, credibility) from specific measures (revenue, expenditure). Because the text is written before the budget year, and the planned numbers are held fixed, any effect of the

wording is information that the numbers do not contain. In answer to the criticism that fiscal surveillance relies on optimistic figures produced by governments, the paper offers a signal that is harder to fake.

To do it, I collect every Draft Budgetary Plan published on the European Commission's website since 2014, extract its text in many languages, and score it against dictionaries[1] for four fiscal concepts. Two general commitments (consolidation and credibility) and two specific measures (revenue-based and expenditure-based). Scores are standardized within each country, so that they capture how unusual a plan's emphasis is relative to the government's own drafting style. Then, I match them to real-time AMECO vintages, which separate the planned change, the first-release outcome and revised data and estimate OLS regressions with country and year fixed effects. The controls (lagged debt, the output gap and elections) are measured with the information available when the plan was written. Because plans are submitted in October of the previous year, the rhetoric is fixed before the outcome is known. Inference relies on a wild cluster bootstrap to account for the small number of countries.

General rhetoric about consolidation and credibility does not predict whether governments deliver their plans, it behaves as cheap talk. In contrast, plans that place more emphasis on specific revenue measures are followed by better than planned structural and cyclically adjusted balances, by about 0.3 to 0.4 percentage points of GDP per standard deviation. This association appears in both first release and revised data and does not depend on when the controls are measured. Language on expenditure restraint carries a weaker signal, limited to the structural balance and about a third the size of the revenue effect. Overall, the specificity of a plan, not its tone, signals commitment, consistent with tax measures being legislated with the budget while spending restraint slips during execution.

The remainder of the paper is organized as follows. Section 2 presents the conceptual framework and its testable hypotheses. Section 3 describes the corpus, the measurement of rhetoric and the fiscal data. Section 4 reports the main results while section5 presents robustness checks. Finally the last section (6) concludes.

2 Conceptual framework

This section presents a simple model of how a government writes its budget plan and then executes it. The model separates two kinds of language. General rhetoric (consolidation, credibility) costs nothing and cannot be verified, so it is cheap talk in the sense of Crawford and Sobel (1982). Specific measures are costly to announce and verifiable once announced. They therefore work as a costly signal Spence (1973), and they bind through audience costs

[1] Dictionaries are available in appendix.

Fearon (1994). The framework yields four hypotheses, which the empirical analysis tests.

2.1 Set-up

A government has a private type $\theta \in \{\theta_L, \theta_H\}$, with $\theta_H > \theta_L > 0$, which measures its willingness to consolidate. In this model, a higher θ means a lower political cost of fiscal effort. As in models of fiscal policy under private information like Rogoff (1990), outsiders (here voters, the Commission and markets) do not observe θ. The type may change over time and within a country with a new government, a coalition change or an approaching election.

The game has two stages, matching the calendar of the Two-Pack.

Stage 1: the plan (October of year $t-1$). The government submits a plan. It contains:

- a planned adjustment a^*, set largely by the fiscal rules and therefore common to both types;
- general rhetoric $g \geq 0$, which is free;
- for each category $j \in \{R, E\}$ (revenue, expenditure), a decision $s_j \in \{0, 1\}$ to specify concrete measures. Doing so has a political cost $k_j > 0$, because naming a tax increase or a spending cut is unpopular.

Let $S = s_R + s_E$ be the number of categories in which measures are specified.

Stage 2: execution (year t). Specified measures deliver part of the adjustment mechanically. A measure of category j contributes κ_j, with

$$\kappa_R > \kappa_E \geq 0. \tag{1}$$

Inequality (1) reflects the different legal nature of the two instruments. A tax measure written into the budget law applies once the law is passed, without further decisions. Spending restraint must be enforced throughout the year, because appropriations are ceilings rather than obligations, and it can be undone by supplementary budgets or by spending revenue windfalls (Beetsma et al., 2022). In addition, expenditure based consolidation (pensions reforms,remove of subsidies) can be hard to implement due to political constraints and immediate costs.

The government then chooses discretionary effort $e \geq 0$, at cost $e^2/(2\theta)$. The delivered adjustment is

$$a \;=\; \kappa_R s_R + \kappa_E s_E + e + \varepsilon, \tag{2}$$

where ε is a macroeconomic shock with mean zero, unknown when the plan is written.

Payoffs. The government values the delivered adjustment at a marginal rate $\mu + cS$. The term $\mu > 0$ captures the general pressure of the fiscal rules and markets. The term cS, with $c > 0$, is the audience cost: a shortfall on a measure that was specified can be verified by the Commission, the national fiscal council and markets, so each specified category raises the reputational value of delivering. General rhetoric does not enter payoffs, since it cannot be verified. The government's payoff is

$$U \;=\; (\mu + cS)\, a \;-\; \frac{e^2}{2\theta} \;-\; k_R s_R \;-\; k_E s_E. \tag{3}$$

This structure, in which precise, verifiable commitments discipline execution, mirrors the "contract" approach to fiscal governance (von Hagen and Harden, 1995; Hallerberg et al., 2009).

2.2 Execution

In stage 2, the first-order condition for effort gives

$$e^*(\theta, S) \;=\; \theta\,(\mu + cS). \tag{4}$$

Effort rises with willingness θ and with the number of specified categories S: announcing measures commits the government to deliver more, because failing to do so is now observable. The expected implementation error, that is the expected gap between delivered and planned adjustment, is

$$\mathbb{E}[a - a^*] \;=\; \kappa_R s_R + \kappa_E s_E + \theta\,(\mu + cS) - a^*. \tag{5}$$

2.3 The plan

In stage 1, the government anticipates e^*. The gain from specifying measures in category j is increasing in θ. Specification raises the marginal value of adjustment by c, and a government with a lower cost of effort draws more value from that higher incentive.[2] This single-crossing property gives the following result.

Proposition 1 *(i) General rhetoric is uninformative. Since g is free and does not affect payoffs, both types can choose the same g, and $\mathbb{E}[a - a^* \mid g]$ does not depend on g.*

[2]Substituting (4) into (3), the value of the plan is $V(\theta, S) = (\mu + cS)\,(\kappa_R s_R + \kappa_E s_E) + \frac{1}{2}\,\theta\,(\mu + cS)^2 - k_R s_R - k_E s_E$. The gain from setting $s_j = 1$ rather than 0 contains the term $\frac{1}{2}\theta\left[(\mu + c(S_0 + 1))^2 - (\mu + cS_0)^2\right]$, where S_0 is the number of other specified categories. This term is strictly increasing in θ, which gives the single-crossing property.

(ii) Let $D_j(\theta)$ be the gain from specifying measures in category j. If $D_j(\theta_L) < k_j \leq D_j(\theta_H)$, there is a separating equilibrium in which only the willing type specifies measures in category j.

(iii) In that equilibrium, the difference in expected delivery between plans that specify category j and plans that do not is

$$\Delta_j = \underbrace{\kappa_j}_{\textit{lock-in}} + \underbrace{c\,\theta_H}_{\textit{commitment}} + \underbrace{(\theta_H - \theta_L)(\mu + cS_0)}_{\textit{selection}} > 0. \tag{6}$$

(iv) Under (1), and with the same commitment and selection terms for both categories, $\Delta_R - \Delta_E = \kappa_R - \kappa_E > 0$.

Equation (6) shows why specific language predicts delivery through three channels:

- *Lock-in*: part of the adjustment is enacted with the budget law.
- *Commitment*: once measures are announced, audience costs raise effort.
- *Selection*: only governments willing to consolidate accept the political cost of naming measures.

The three channels act together, and the empirical analysis does not try to separate them. All three imply that the text contains information about delivery that the planned number a^* does not.

2.4 Testable hypotheses

The empirical counterparts of the model are direct:

- the within-country z-scores of consolidation and credibility rhetoric measure g;
- the z-scores of revenue- and expenditure-based rhetoric measure s_R and s_E;
- the implementation error (9) corresponds to $a - a^*$, with the planned change controlled for;
- country fixed effects absorb each country's average willingness to consolidate, so the coefficients are identified from changes in θ within a country.

Because the plan is written before ε is realised, the model predicts no correlation between the text and the shock. The four hypotheses below follow (Table 1).

H1 (cheap talk). General consolidation and credibility rhetoric does not predict the implementation error: $\beta_g = 0$.

Table 1: Hypotheses, model predictions and empirical tests

Hypothesis	Model prediction	Empirical test
H1 Cheap talk	Proposition 1(i)	$\beta = 0$ for consolidation and credibility z-scores, headline and cyclically adjusted balances
H2 Costly signal	Proposition 1(iii): $\Delta_j > 0$	$\beta > 0$ for revenue- and expenditure-based z-scores; implementation error and first-release change
H3 Lock-in asymmetry	Proposition 1(iv): $\Delta_R > \Delta_E$	$\beta_R > \beta_E$; Wald test of equality in specification S2
H4 Policy, not cycle	Text is fixed before ε	Same sign and similar size for the structural, cyclically adjusted and primary balances

H2 (costly signal). Specific measures predict better-than-planned delivery: $\beta_{s_j} > 0$.

H3 (lock-in asymmetry). Revenue measures predict delivery more strongly than expenditure measures: $\beta_{s_R} > \beta_{s_E}$.

H4 (policy, not cycle). The effect of specific measures appears in the structural and cyclically adjusted balances, not only in the headline balance, because it operates through policy rather than through the shock ε.

The model also predicts that the effect of specific measures should increase with the audience cost c, since $\partial\Delta_j/\partial c = \theta_H > 0$. Episodes of stronger scrutiny (high sovereign spreads, a warning letter from the Commission, a strong national fiscal council) should therefore display larger coefficients. This prediction is left for future work.

3 Empirical strategy

This section describes how the published budget documents are turned into country-year measures of fiscal rhetoric and how these measures are matched to real-time fiscal outcomes. The procedure has five steps, summarised in Figure 1: document collection, text extraction, dictionary scoring, within-country standardisation, and matching with real-time fiscal data. Each step produces an intermediate file, so that every score can be traced back to the phrases that generated it.

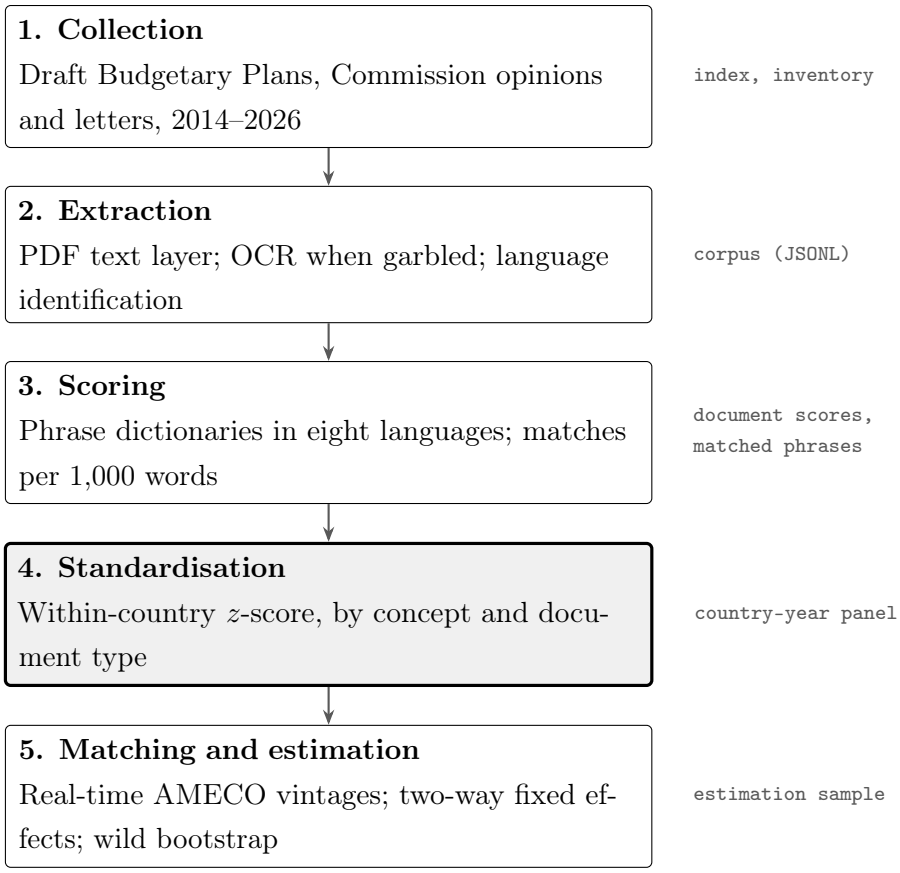


Figure 1: From published documents to the estimation sample

Notes: Each step writes an intermediate file (right), so that the scores can be audited and the steps re-run independently.

3.1 The corpus of Draft Budgetary Plans

Since the entry into force of the Two-Pack (European Parliament and Council of the European Union, 2013), every euro-area member state submits a Draft Budgetary Plan (DBP) for the following year to the European Commission and the Eurogroup by 15 October. The Commission assesses each plan against the requirements of the Stability and Growth Pact and publishes an opinion, normally by the end of November. The plans are the earliest standardised statement of a government's fiscal intentions for the coming year: they are produced on a common calendar and a common template, and they are submitted before the national budget is adopted. This makes them well suited to a cross-country comparison of fiscal communication.

I collect the full set of documents published on the Commission's DBP pages, which list the submissions for each budget year from 2014 to 2026. A scraper visits each yearly page, assigns every file to a member state (from the document title, the page section and, failing

those, the file name), and classifies it into one of seven types: the original plan, an updated or resubmitted plan, the Commission opinion, the accompanying staff working document, a letter from the Commission requesting clarification, the government's reply, and the Commission's overall assessment of the euro area. Only files that are genuine PDF documents are kept, and no file is ever overwritten, so the corpus can be extended incrementally. Where a plan was missing from the Commission's pages, mostly for early years or for national-language versions, I added the official document obtained from the national ministry of finance. These documents are flagged throughout, and the main results are re-estimated without them.

The plans and opinions are the documents analysed as text. Letters and resubmissions are not scored. Instead, they define two indicators: whether the Commission wrote to the government before issuing its opinion, and whether the plan was resubmitted. Both signal that the Commission considered the original plan deficient.

3.2 Text extraction and language identification

Text is extracted from the PDF text layer. Some plans either have no text layer or embed fonts without a character map, so that extraction returns unreadable output. A document is therefore sent to optical character recognition (Tesseract; Smith, 2007) in its country's language whenever extraction fails or the extracted text is garbled. Text is treated as garbled when it contains more than 20 unmapped glyph tokens, when replacement characters exceed 1% of characters, or when letters account for less than 45% of characters.

The submission language varies: most governments file in English, others in their national language or in both. The language of each document is identified from the density of short function words specific to that language, computed after removing accents. A language is assigned when its markers exceed 6% of tokens. Documents shorter than 2,000 characters, typically cover pages or failed extractions, are discarded.

When the same document exists in several languages for a given country-year, a single primary version is retained. This is the English version when available, and otherwise the national-language one. All subsequent measures are computed on primary documents only.

3.3 Measuring fiscal rhetoric

I measure fiscal rhetoric with a dictionary method (Loughran and McDonald, 2011; Baker et al., 2016; Gentzkow et al., 2019). Each document is scored against dictionaries written for four concepts, summarised in Table 2. Two concepts capture *general* commitments: *consolidation* covers deficit and debt reduction and references to the fiscal rules, and *credibility* covers language asserting that targets will be met. Two capture *specific* measures: *revenue-based*

Table 2: Fiscal concepts and example dictionary entries

Concept	Content	Example entries (English dictionary)
Consolidation	Deficit and debt reduction; fiscal rules and targets	*fiscal consolidation, structural balance, medium-term objective, debt brake*
Credibility	Statements that targets will be met; institutional safeguards	*credible, track record, fiscal council, prudent forecast, slippage*
Revenue-based	Identifiable tax and revenue measures	*tax base, VAT rate, tax evasion, social security contributions, windfall tax*
Expenditure-based	Identifiable spending restraint	*spending review, wage bill, hiring freeze, expenditure ceiling, pension reform*

Notes: Dictionaries exist in English, French, Spanish, German, Italian, Portuguese, Dutch and Slovak and are written for each language, not translated word for word. Entries are phrases. A trailing wildcard matches inflected forms. The full lists are reported in the Online Appendix.

language refers to identifiable tax and revenue instruments, and *expenditure-based* language to identifiable spending restraint. The distinction between general commitments and specific measures is the core of the empirical design. General commitments cost little to write, whereas specific measures are verifiable and expose the government if they are not implemented.

The dictionaries follow three rules designed to limit measurement error (Grimmer and Stewart, 2013). First, entries are multi-word phrases wherever possible, which reduces ambiguity relative to single words. Second, descriptors that appear throughout every budget document, such as "public debt" or "net borrowing", are excluded, because they would dominate the counts without distinguishing one plan from another. Third, no phrase belongs to more than one concept, which is verified automatically. Documents and dictionary entries are normalised in the same way: text is lower-cased, accents and ligatures are removed, and digits and punctuation are replaced by spaces. Matching is therefore insensitive to accents, hyphenation and apostrophes in all languages.

The raw score of concept c in document d is the number of dictionary matches per thousand words:

$$s_{c,d} \;=\; 1000 \times \frac{m_{c,d}}{N_d}, \tag{7}$$

where $m_{c,d}$ is the number of matches and N_d the number of words in the document. Normalising by length is necessary because the plans range from short notes to documents of more than a hundred pages. The full list of matched phrases is retained, so that the contribution of each entry to each score can be inspected.

3.4 Within-country standardisation

Raw frequencies are not comparable across countries. Drafting traditions differ, some plans are written in English while others are translated, and the dictionaries differ in coverage across languages. Rather than compare levels across countries, I measure how unusual a plan's emphasis is relative to the same government's other plans. Each score is standardised within country i and document type, over all budget years t in which a primary document is available:

$$z_{c,it} \;=\; \frac{s_{c,it} - \bar{s}_{c,i}}{\hat{\sigma}_{c,i}}, \tag{8}$$

where $\bar{s}_{c,i}$ and $\hat{\sigma}_{c,i}$ are the country-specific mean and standard deviation of the score. The z-score is set to missing when a country has fewer than three plans or when the concept shows no variation for that country. In the current sample, for example, no German plan contains a credibility entry.

The standardisation has two advantages. It removes level differences that originate in language and drafting style, which would otherwise load onto the country fixed effects in an uninformative way. It also gives the coefficients a direct interpretation: the effect of a plan that is one standard deviation more focused on a given concept than is usual for that government. The same procedure is applied to the Commission's opinions and staff working documents, which allows a comparison of what governments emphasise with what the Commission emphasises.

3.5 Real-time fiscal outcomes

Fiscal data come from the European Commission's AMECO database (European Commission, 2026). Following the real-time fiscal literature (Beetsma et al., 2009; Frankel and Schreger, 2013; Cimadomo, 2016), I use archived vintages to distinguish what was expected when the plan was assessed from what was eventually recorded (Table 3). For each fiscal variable x, the planned change $\Delta x_{it}^{\text{plan}}$ for budget year t is taken from the autumn vintage of year $t-1$, published in the same weeks as the plans and the Commission's opinions.[3] The first-release change $\Delta x_{it}^{\text{first}}$ is taken from the spring vintage of year $t+1$, the first in which year-t outturns are reported, and the latest change $\Delta x_{it}^{\text{latest}}$ from the current release.

The implementation error is defined as

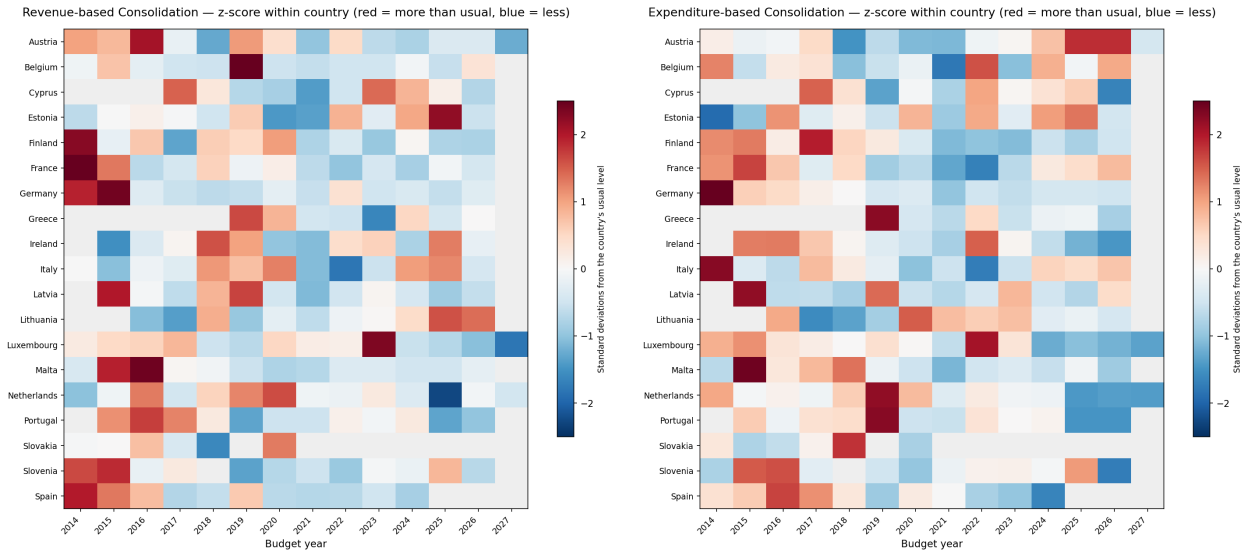
$$e_{it}^{x} \;=\; \Delta x_{it}^{\text{first}} - \Delta x_{it}^{\text{plan}}, \tag{9}$$

[3]The autumn forecast is the Commission's projection. It incorporates the measures announced in the Draft Budgetary Plan but may differ from the targets the government states. As a robustness check, the planned change is replaced by the government's own targets as reported in each plan.

Table 3: Data vintages used to measure plans and outcomes

Variable	AMECO vintage	Interpretation
$\Delta x_{it}^{\text{plan}}$	Autumn, year $t-1$	Change expected when the plan is assessed
$\Delta x_{it}^{\text{first}}$	Spring, year $t+1$	First official outturn
$\Delta x_{it}^{\text{latest}}$	Current release	Revised outturn

so that a positive value for a balance indicates that the government did better than planned. The outcome variables x are the headline general government balance, the cyclically adjusted balance, the cyclically adjusted primary balance, the structural balance, gross debt, total revenue and total expenditure, all in percent of GDP. Control variables are lagged gross debt, the output gap and an indicator for national election years, constructed from the ParlGov database (Döring and Manow, 2026). Real GDP growth, inflation and the lagged long-term interest rate, from AMECO, Eurostat and the European Central Bank, are used in additional specifications for debt. The next graphs (2a and 2b) represent the evolution of z scores for specific measures through the time.



(a) Revenue-based measures

(b) Expenditure-based measures

Figure 2: Emphasis on specific measures in the Draft Budgetary Plans, by country and year

Notes: Within-country z-scores of the dictionary scores of each plan.

3.6 Empirical strategy

To examine whether fiscal-plan rhetoric contains information about subsequent fiscal outcomes beyond that conveyed by planned fiscal adjustments, I estimate a country-year panel regression with country and budget-year fixed effects. The baseline specification is :

$$Y_{it} = \boldsymbol{\beta}'\mathbf{z}_{it} + \gamma\Delta x_{it}^{\text{plan}} + \delta\text{Debt}_{i,t-1} + \theta\text{OG}_{it} + \lambda\text{Election}_{it} + \alpha_i + \tau_t + \varepsilon_{it}, \tag{10}$$

where i indexes countries and t denotes the budget year covered by the fiscal plan. Y_{it} is the fiscal outcome of interest, $\mathbf{z}_{it}$ is a vector of standardised rhetoric measures, and $\Delta x_{it}^{\text{plan}}$ denotes the planned change in the corresponding fiscal variable. The specification also controls for lagged gross debt, the output gap, and an election-year indicator. Country fixed effects (α_i) account for time-invariant country-specific characteristics, while budget-year fixed effects (τ_t) absorb shocks and institutional changes common to all countries in a given year.

The vector $\mathbf{z}_{it}$ captures the emphasis placed on the fiscal concepts identified in the plans. Specification S1 includes consolidation rhetoric alone, whereas specification S2 includes all four concepts simultaneously, allowing the association of each concept with fiscal outcomes to be estimated conditional on the other rhetorical dimensions. The rhetoric measures are standardised as defined in equation (8). Consequently, their coefficients measure the change in the outcome, in percentage points of GDP, associated with a one-standard-deviation increase in the emphasis placed on a given concept.

The analysis considers three measures of fiscal performance for each of the seven fiscal variables described in Section 3.5. The first-released change in the fiscal variable, $\Delta x_{it}^{\text{first}}$; the implementation error, e_{it}^{x}, defined in equation (9); and the latest available change, $\Delta x_{it}^{\text{latest}}$. Each outcome is estimated separately. When the dependent variable is a realised fiscal change, the planned change is included as a control. Its coefficient, γ, measures the conditional association between the planned adjustment and the realised change. When the dependent variable is the implementation error, the planned change is excluded because it is already incorporated into the construction of the error.

The timing of fiscal-plan preparation helps establish the temporal ordering of the analysis. The plan covering budget year t is drafted during the summer of year $t-1$ and submitted by 15 October of year $t-1$. The rhetoric measures and planned fiscal changes are therefore observed before the corresponding budget is executed and its fiscal outcomes are realised. The first release of the fiscal outturn typically occurs in spring of year $t+1$. This ordering rules out reverse causality from realised budget-year outcomes to the wording of the previously submitted plan.

The country and year fixed effects identify the coefficients from within-country changes over time, net of changes common to all countries. Accordingly, β_c measures the conditional association between a one-standard-deviation increase in rhetoric about concept c and the subsequent fiscal outcome. When the dependent variable is the implementation error, a positive coefficient indicates that stronger emphasis on the concept is associated with a more favourable outcome relative to the plan, conditional on the included controls.

The estimates are interpreted as predictive conditional associations rather than strictly causal effects. Although the rhetoric measures precede the realised outcomes, governments may possess information when drafting their plans that affects both the language used and subsequent fiscal performance. To the extent that this information is not captured by the planned fiscal figures and other controls, the estimated coefficients may reflect this additional information. This interpretation is consistent with the paper's objective of assessing whether fiscal-plan rhetoric contains information about subsequent implementation beyond that conveyed by planned fiscal adjustments.

Figure 3 summarises this sequence and shows which data release each variable comes from, in the baseline and in the robustness checks. The regressors of interest are thus predetermined with respect to the outcome. They cannot respond to shocks that occur during the budget year, so reverse causality from realised outcomes to the wording of the plan is ruled out by construction.

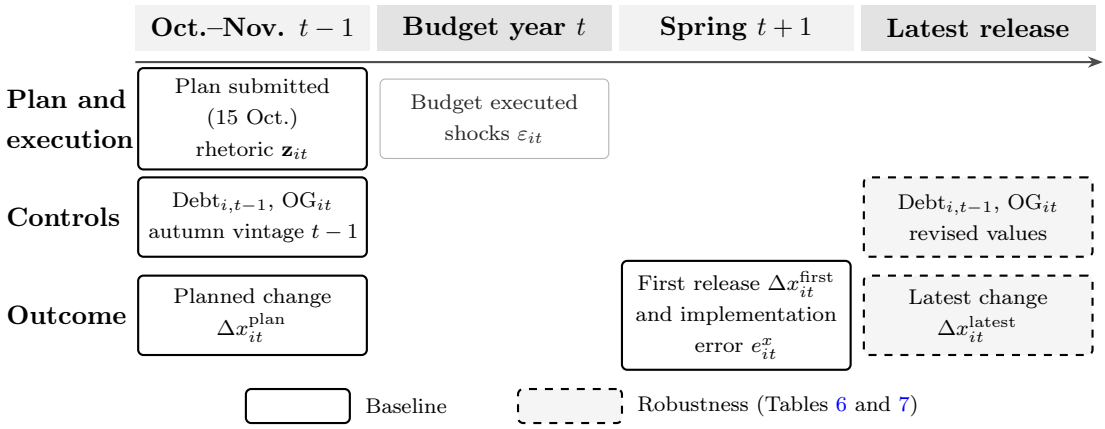


Figure 3: Timing of the plan, the controls and the outcomes

Notes: Variables are indexed by the budget year t that the plan covers. The plan, its rhetoric, the planned change and the baseline controls all date from the autumn of year $t-1$, before the budget year starts. The outcome is first measured in the spring vintage of year $t+1$. Robustness checks replace the outcome or the controls with their values in the latest AMECO release, after all revisions.

4 Results

The results are consistent with the model's four predictions. General rhetoric is cheap talk. In specification S2, the consolidation and credibility scores never predict the implementation error or the first-release change of any of the four balances (Tables 4 and 5), as implied by Proposition 1(i) and H1. Consolidation rhetoric is associated with a better structural balance when it enters alone (Panel A), but this association disappears once the specific measures are controlled for (Panel B), suggesting that it reflected the measures that accompany

consolidation language rather than the language itself.

Specific measures, by contrast, carry real information. A one-standard-deviation increase in revenue-based rhetoric improves the implementation error of the structural balance by 0.324 percentage points of GDP ($p < 0.01$), of the cyclically adjusted balance by 0.370 points ($p < 0.05$) and of the cyclically adjusted primary balance by 0.362 points ($p < 0.1$), and raises the revenue outturn by 0.248 points relative to plan ($p < 0.1$) (H2). The structural-balance effect is also present in first-release changes (0.260, $p < 0.05$). That the effects are strongest for the structural and cyclically adjusted balances, rather than for the headline balance, is consistent with an effect that operates through policy rather than through the shock ε (H4).

The asymmetry between the two categories is the central result. Revenue-based rhetoric is associated with better outcomes for all cyclically adjusted measures of the balance. Expenditure-based rhetoric is associated with a better structural balance (0.124, $p < 0.1$) and a lower debt ratio (-0.602, $p < 0.1$), but its coefficients are about a third the size of the revenue coefficients, and it barely moves expenditure (0.017, column 7). This is the pattern implied by the lock-in inequality $\kappa_R > \kappa_E$ (H3): tax measures take effect once the budget law is passed, whereas spending restraint must be enforced throughout the year. Pooling the two categories into a single index would average away this asymmetry; specific measures should therefore not be treated uniformly. Rhetoric matters when it is specific, and it matters most on the revenue side.

Two caveats apply. The coefficients combine the lock-in, commitment and selection channels of equation (6), which the design does not separate; and specific plans may partly reveal private information rather than cause delivery. With 18 country clusters and several outcomes, individual coefficients significant at the 10% level should also be read as part of the overall pattern rather than in isolation. Neither caveat, however, alters the qualitative message. The specificity of plan language, and the side of the budget it commits to, carry systematic information about what will be delivered.

Table 4: Plan rhetoric and the implementation error

	Budget balance					Components	
	Headline	CAPB	CAB	Structural	Debt	Revenue	Expenditure
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Consolidation rhetoric only (S1)							
Consolidation (z)	−0.097	−0.083	−0.040	0.185**	0.138	−0.049	0.047
	(0.156)	(0.165)	(0.153)	(0.076)	(0.243)	(0.078)	(0.177)
Observations	162	162	162	162	162	162	162
Countries	18	18	18	18	18	18	18
Panel B. All rhetoric measures (S2)							
Consolidation (z)	−0.134	−0.108	−0.087	0.056	0.452	−0.114	0.019
	(0.118)	(0.135)	(0.125)	(0.084)	(0.350)	(0.112)	(0.155)
Credibility (z)	−0.162	−0.065	−0.079	0.025	−0.150	−0.058	0.103
	(0.124)	(0.150)	(0.151)	(0.098)	(0.316)	(0.069)	(0.158)
Revenue measures (z)	0.287	0.362*	0.370**	0.324***	0.116	0.248*	−0.040
	(0.176)	(0.179)	(0.173)	(0.099)	(0.335)	(0.118)	(0.191)
Expenditure measures (z)	0.049	−0.054	−0.006	0.124*	−0.602*	0.068	0.017
	(0.142)	(0.126)	(0.131)	(0.067)	(0.315)	(0.101)	(0.140)
Observations	162	162	162	162	162	162	162
Countries	18	18	18	18	18	18	18
Planned change	No	No	No	No	No	No	No
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: OLS estimates of equation (10). The dependent variable is the implementation error, $e^x_{it} = \Delta x^{\text{first}}_{it} - \Delta x^{\text{plan}}_{it}$: the first-release change in year t (spring vintage of $t+1$) minus the planned change (autumn vintage of $t-1$). A positive value means a better-than-planned outcome for the balances, and a larger-than-planned increase for debt, revenue and expenditure. Each column is a separate regression; outcomes are in percentage points of GDP. Rhetoric variables are within-country z-scores of the plan's dictionary scores, so coefficients are the change in the outcome associated with a one-standard-deviation increase. Controls are measured with the information available when the plan was written: gross debt in $t-1$ and the output gap in t, both from the autumn vintage of $t-1$, and an election-year indicator. The sample covers budget years 2014–2023 (10 years). Standard errors clustered by country in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: Plan rhetoric and the realised change (first release)

	Budget balance					Components	
	Headline	CAPB	CAB	Structural	Debt	Revenue	Expenditure
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Consolidation rhetoric only (S1)							
Consolidation (z)	−0.154	−0.088	−0.054	0.175**	0.133	−0.060	0.094
	(0.147)	(0.160)	(0.150)	(0.072)	(0.241)	(0.069)	(0.163)
Observations	162	162	162	162	162	162	162
Countries	18	18	18	18	18	18	18
Panel B. All rhetoric measures (S2)							
Consolidation (z)	−0.174	−0.109	−0.097	0.058	0.462	−0.109	0.063
	(0.111)	(0.127)	(0.117)	(0.073)	(0.356)	(0.083)	(0.150)
Credibility (z)	−0.155	−0.058	−0.065	0.029	−0.184	−0.029	0.117
	(0.132)	(0.148)	(0.150)	(0.095)	(0.338)	(0.071)	(0.152)
Revenue measures (z)	0.203	0.277	0.289*	0.260**	0.124	0.129	−0.041
	(0.164)	(0.169)	(0.161)	(0.096)	(0.344)	(0.089)	(0.186)
Expenditure measures (z)	0.047	−0.033	0.013	0.123**	−0.619*	0.065	0.019
	(0.125)	(0.109)	(0.112)	(0.056)	(0.314)	(0.085)	(0.132)
Observations	162	162	162	162	162	162	162
Countries	18	18	18	18	18	18	18
Planned change	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: OLS estimates of equation (10). The dependent variable is the first-release change $\Delta x_{it}^{\text{first}}$, from the spring vintage of year $t+1$. The planned change $\Delta x_{it}^{\text{plan}}$ is included as a control. Each column is a separate regression; outcomes are in percentage points of GDP. Rhetoric variables are within-country z-scores of the plan's dictionary scores, so coefficients are the change in the outcome associated with a one-standard-deviation increase. Controls are measured with the information available when the plan was written: gross debt in $t-1$ and the output gap in t, both from the autumn vintage of $t-1$, and an election-year indicator. The sample covers budget years 2014–2023 (10 years). Standard errors clustered by country in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

5 Robustness checks

5.1 Use of latest fiscal data

Table 6: Plan rhetoric and the realised change (latest data)

	Budget balance					Components	
	Headline	CAPB	CAB	Structural	Debt	Revenue	Expenditure
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Consolidation rhetoric only (S1)							
Consolidation (z)	−0.053	0.006	0.041	0.157	0.050	−0.079	−0.019
	(0.118)	(0.134)	(0.125)	(0.111)	(0.224)	(0.060)	(0.135)
Observations	179	179	179	179	179	179	179
Countries	18	18	18	18	18	18	18
Panel B. All rhetoric measures (S2)							
Consolidation (z)	−0.059	0.006	0.025	0.144	0.301	−0.074	−0.014
	(0.103)	(0.137)	(0.126)	(0.110)	(0.326)	(0.083)	(0.135)
Credibility (z)	−0.141	−0.068	−0.068	−0.106	−0.455	−0.097	0.034
	(0.126)	(0.161)	(0.169)	(0.142)	(0.373)	(0.081)	(0.160)
Revenue measures (z)	0.164	0.252	0.255	0.318**	0.322	0.111	−0.009
	(0.135)	(0.171)	(0.159)	(0.139)	(0.389)	(0.091)	(0.171)
Expenditure measures (z)	0.012	−0.077	−0.046	−0.057	−0.418	−0.008	−0.024
	(0.106)	(0.100)	(0.103)	(0.092)	(0.294)	(0.088)	(0.124)
Observations	179	179	179	179	179	179	179
Countries	18	18	18	18	18	18	18
Planned change	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: OLS estimates of equation (10). The dependent variable is the change $\Delta x_{it}^{\text{latest}}$ in the latest AMECO release. The planned change $\Delta x_{it}^{\text{plan}}$ is included as a control. Each column is a separate regression; outcomes are in percentage points of GDP. Rhetoric variables are within-country z-scores of the plan's dictionary scores, so coefficients are the change in the outcome associated with a one-standard-deviation increase. Controls are measured with the information available when the plan was written: gross debt in $t-1$ and the output gap in t, both from the autumn vintage of $t-1$, and an election-year indicator. The sample covers budget years 2014–2024 (11 years). Standard errors clustered by country in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

As a robustness check, I re-estimate the baseline specification using the latest available AMECO data instead of the first available estimates of fiscal outcomes. This allows me to assess whether the results are sensitive to subsequent statistical revisions, including those arising from GDP reassessments and updated fiscal data. The specification remains unchanged, controlling for the planned fiscal adjustment and other macro-fiscal factors, with country and year fixed effects.

The results are still consistent with the baseline findings. General consolidation rhetoric and credibility-related language remain statistically insignificant across all outcomes. By contrast, revenue-based language is positively associated with changes in the structural balance. A one-standard-deviation increase in revenue-related rhetoric is associated with an approximately 0.32 percentage-point improvement in the structural balance, conditional on the planned change and other controls. The coefficient is statistically significant at the 5% level, while the estimates for the other fiscal indicators are not statistically significant. These findings suggest that the relationship between specific revenue measures and fiscal outcomes is not driven solely by the use of first-release data. This supports the interpretation that the specificity of budgetary plans may signal fiscal commitment.

5.2 Timing of the control variables

The baseline controls for the fiscal and cyclical position using the information available when the plan was written. To assess whether this choice matters, I re-estimate the model with the controls measured in two alternative vintages:

$$Y_{it} \;=\; \boldsymbol{\beta}'\mathbf{z}_{it} + \gamma\,\Delta x_{it}^{\text{plan}} + \delta\,\text{Debt}_{i,t-1}^{(v)} + \theta\,\text{OG}_{it}^{(v)} + \lambda\,\text{Election}_{it} + \alpha_i + \tau_t + \varepsilon_{it}, \qquad v \in \{\text{plan}, \text{latest}\}, \tag{11}$$

where $v = \text{plan}$ denotes the autumn vintage of year $t-1$, released together with the plan, and $v = \text{latest}$ the current AMECO release.

Only the measurement of debt and of the output gap changes; the dependent variable, the rhetoric measures, the planned change and the sample are the same in both cases. The plan-time vintage is the preferred specification, because it contains only what the government knew when drafting the plan, and the output gap forecast is not affected by shocks that occur during the budget year. The latest vintage is more accurate but incorporates later revisions, and the realised output gap partly reflects the fiscal outcome itself. Table 7 shows that the association between revenue-based rhetoric and the structural balance is virtually unchanged across the two vintages (0.32 and 0.30 for the implementation error, both significant). The effect on the headline balance, by contrast, becomes significant only with latest-vintage controls. The structural result therefore does not depend on how the cyclical position is

measured, whereas the headline result is sensitive to it.

Table 7: Robustness to the timing of the control variables

	Headline		CAPB		Structural		Debt	
	Plan	Latest	Plan	Latest	Plan	Latest	Plan	Latest
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Implementation error								
Consolidation (z)	−0.134	−0.155	−0.108	−0.117	0.056	0.065	0.452	0.441
	(0.118)	(0.127)	(0.135)	(0.127)	(0.084)	(0.085)	(0.350)	(0.289)
Credibility (z)	−0.162	−0.190	−0.065	−0.080	0.025	0.011	−0.150	−0.068
	(0.124)	(0.125)	(0.150)	(0.140)	(0.098)	(0.095)	(0.316)	(0.298)
Revenue measures (z)	0.287	0.386**	0.362*	0.340*	0.324***	0.296**	0.116	−0.216
	(0.176)	(0.170)	(0.179)	(0.187)	(0.099)	(0.125)	(0.335)	(0.348)
Expenditure measures (z)	0.049	0.069	−0.054	−0.028	0.124*	0.110	−0.602*	−0.526**
	(0.142)	(0.116)	(0.126)	(0.104)	(0.067)	(0.068)	(0.315)	(0.247)
Panel B. Realised change (first release)								
Consolidation (z)	−0.174	−0.187	−0.109	−0.117	0.058	0.068	0.462	0.446
	(0.111)	(0.121)	(0.127)	(0.125)	(0.073)	(0.077)	(0.356)	(0.289)
Credibility (z)	−0.155	−0.179	−0.058	−0.069	0.029	0.018	−0.184	−0.134
	(0.132)	(0.128)	(0.148)	(0.141)	(0.095)	(0.096)	(0.338)	(0.301)
Revenue measures (z)	0.203	0.293*	0.277	0.267	0.260**	0.243*	0.124	−0.196
	(0.164)	(0.153)	(0.169)	(0.171)	(0.096)	(0.116)	(0.344)	(0.349)
Expenditure measures (z)	0.047	0.058	−0.033	−0.015	0.123**	0.104*	−0.619*	−0.530**
	(0.125)	(0.100)	(0.109)	(0.088)	(0.056)	(0.056)	(0.314)	(0.233)
Observations	162	162	162	162	162	162	162	162
Countries	18	18	18	18	18	18	18	18

Notes: OLS estimates of equation (10), specification S2, on a common sample. Columns "Plan" use controls measured when the plan was written (debt in $t-1$ and the output gap in t from the autumn vintage of $t-1$), as in the baseline. Columns "Latest" replace them with the same variables from the latest AMECO release, which incorporate later revisions and information not available to the government. Both include an election-year indicator, country and year fixed effects, and, in Panel B, the planned change. Standard errors clustered by country in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

5.3 Difference of coefficient

The theoretical framework suggests that revenue-based measures have larger coefficients than expenditure-based measures. To assess whether this difference reflects a systematic

asymmetry rather than sampling noise, I test whether the two coefficients differ. For each balance and outcome definition, I re-estimate specification S2 and calculate the difference $\hat{\beta}_R - \hat{\beta}_E$. Standard errors are clustered by country and account for the covariance between the two estimates.

Since hypothesis H3 predicts a positive difference, I test $H_0 \colon \beta_R \leq \beta_E$ against $\beta_R > \beta_E$. Table 8 presents the results. The difference is positive in all twelve specifications and statistically significant at the 10% level in eight. At the 5% level, the difference is significant for the implementation error of the cyclically adjusted primary balance, the cyclically adjusted balance, and the structural balance, with estimates ranging from 0.20 to 0.42 percentage points of GDP. The difference is also significant for the structural balance using the latest data.

The headline balance is the only measure for which the difference is never statistically significant, as neither coefficient is precisely estimated.

Overall, the results suggest that the asymmetry between revenue and expenditure measures is not simply due to imprecise estimates. Revenue measures are more consistently associated with better policy delivery than expenditure measures, in line with the model's lock-in mechanism.

Table 8: Revenue versus expenditure measures: test of equal coefficients

	Implementation error	Realised change (first release)	Realised change (latest data)
	(1)	(2)	(3)
Headline balance	0.238	0.156	0.152
	(0.242)	(0.228)	(0.172)
	[0.170]	[0.252]	[0.196]
CAPB	0.416**	0.310*	0.329*
	(0.214)	(0.198)	(0.201)
	[0.034]	[0.068]	[0.060]
Cyclically adjusted balance	0.375**	0.277*	0.301*
	(0.215)	(0.193)	(0.190)
	[0.049]	[0.085]	[0.066]
Structural balance	0.200**	0.138	0.375**
	(0.110)	(0.107)	(0.177)
	[0.043]	[0.108]	[0.024]
Observations	162	162	179
Countries	18	18	18

Notes: Each cell reports the difference $\hat{\beta}_R - \hat{\beta}_E$ between the coefficients on revenue-based and expenditure-based rhetoric in specification S2 of equation (10), estimated by OLS with country and year fixed effects and the baseline controls. Standard errors of the difference, clustered by country and accounting for the covariance between the two coefficients, are in parentheses. One-sided p-values for $H_0 \colon \beta_R \leq \beta_E$ against H3 ($\beta_R > \beta_E$) are in brackets, computed from a t distribution with $G - 1$ degrees of freedom. ** $p < 0.05$, * $p < 0.1$ (one-sided).

6 Conclusion

This paper examines whether the language used in Draft Budgetary Plans submitted by euro-area governments is associated with subsequent fiscal outcomes and budget implementation. The results suggest that general commitments to fiscal consolidation and credibility do not systematically translate into better fiscal performance. By contrast, more specific revenue-based language is positively associated with subsequent improvements in the structural and cyclically adjusted balances, including implementation outcomes that are more favourable than initially planned. These findings remain broadly consistent when using revised fiscal data and alternative timings for macro-fiscal controls. Overall, the evidence suggests that the specificity of fiscal plans may signal governments' commitment to fiscal adjustment, whereas

general rhetorical commitments alone provide limited information about their subsequent delivery.

These findings contribute to the literature on fiscal credibility by highlighting the importance of the content and specificity of fiscal communication. While previous approaches often focus on fiscal targets and planned adjustments, the results suggest that the language used to describe the measures supporting these plans also contains relevant information about their subsequent implementation. In particular, distinguishing between general commitments to consolidation and references to specific revenue measures helps identify which forms of fiscal rhetoric are more closely associated with actual budgetary outcomes. This suggests that the credibility of fiscal plans may depend not only on the ambition of announced targets, but also on how concretely governments explain the measures intended to achieve them. More broadly, the paper highlights the potential value of analyzing budgetary texts alongside conventional fiscal indicators to better understand the relationship between policy intentions and fiscal outcomes.

These findings suggest a concrete role for textual analysis in European fiscal surveillance. The European Commission could systematically assess whether Draft Budgetary Plans translate consolidation objectives into clearly specified and credible measures. Particular attention should be paid to expenditure-based measures (pensions reforms public sector bills, subsidies reforms, etc), whose implementation may depend on concrete spending decisions and administrative follow-through. However, the present results do not establish that expenditure measures are intrinsically more difficult to implement, and the estimated associations do not provide direct evidence of their implementation constraints. Rather, they support examining the specificity of expenditure commitments and monitoring whether announced measures are reflected in subsequent fiscal outcomes. More broadly, combining textual indicators with quantitative fiscal projections could help the Commission identify plans that warrant closer examination, without treating rhetoric alone as proof of policy credibility or implementation.

References

Alesina, A., Favero, C., and Giavazzi, F. (2015). The output effect of fiscal consolidation plans. *Journal of International Economics*, 96(Supplement 1):S19–S42.

Alesina, A., Favero, C., and Giavazzi, F. (2019). *Austerity: When It Works and When It Doesn't*. Princeton University Press, Princeton, NJ.

Baker, S. R., Bloom, N., and Davis, S. J. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4):1593–1636.

Beetsma, R., Bluhm, B., Giuliodori, M., and Wierts, P. (2011). From first-release to ex-post fiscal data: Exploring the sources of revision errors in the EU. CEPR Discussion Paper 8413, Centre for Economic Policy Research.

Beetsma, R., Busse, M., Germinetti, L., Giuliodori, M., and Larch, M. (2022). Is the road to hell paved with good intentions? An empirical analysis of budgetary follow-up in the EU. CEPR Discussion Paper 17154, Centre for Economic Policy Research.

Beetsma, R., Giuliodori, M., and Wierts, P. (2009). Planning to cheat: EU fiscal policy in real time. *Economic Policy*, 24(60):753–804.

Brück, T. and Stephan, A. (2006). Do eurozone countries cheat with their budget deficit forecasts? *Kyklos*, 59(1):3–15.

Cimadomo, J. (2016). Real-time data and fiscal policy analysis: A survey of the literature. *Journal of Economic Surveys*, 30(2):302–326.

Crawford, V. P. and Sobel, J. (1982). Strategic information transmission. *Econometrica*, 50(6):1431–1451.

Döring, H. and Manow, P. (2026). ParlGov: Parliaments and governments database. Development version, https://www.parlgov.org.

Efstathiou, K. and Wolff, G. B. (2019). What drives implementation of the European Union's policy recommendations to its member countries? CEPR Discussion Paper 14032, Centre for Economic Policy Research.

European Commission (2026). AMECO: Annual macro-economic database of the european commission's directorate general for economic and financial affairs. Current release and archived vintages.

European Parliament and Council of the European Union (2013). Regulation (EU) no 473/2013 of 21 may 2013 on common provisions for monitoring and assessing draft budgetary plans and ensuring the correction of excessive deficit of the member states in the euro area. Official Journal of the European Union, L 140, 27.5.2013, pp. 11–23.

Evdokimova, T. and Imam, P. A. (2026). The fiscal communication paradox: Many voices, one budget. IMF Working Paper 26/117, International Monetary Fund. Title as in the SUERF Policy Brief No. 1532 summarising the paper; check the exact working-paper title.

Fearon, J. D. (1994). Domestic political audiences and the escalation of international disputes. *American Political Science Review*, 88(3):577–592.

Frankel, J. and Schreger, J. (2013). Over-optimistic official forecasts and fiscal rules in the eurozone. *Review of World Economics*, 149(2):247–272.

Gentzkow, M., Kelly, B., and Taddy, M. (2019). Text as data. *Journal of Economic Literature*, 57(3):535–574.

Grimmer, J. and Stewart, B. M. (2013). Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3):267–297.

Hallerberg, M., Strauch, R. R., and von Hagen, J. (2009). *Fiscal Governance in Europe*. Cambridge University Press, Cambridge.

Jonung, L. and Larch, M. (2006). Improving fiscal policy in the EU: The case for independent forecasts. *Economic Policy*, 21(47):491–534.

Larch, M., Malzubris, J., and Busse, M. (2021). Optimism is bad for fiscal outcomes. VoxEU.org column, Centre for Economic Policy Research.

Latifi, A., Naboka-Krell, V., Tillmann, P., and Winker, P. (2024). Fiscal policy in the Bundestag: Textual analysis and macroeconomic effects. *European Economic Review*, 168.

Loughran, T. and McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1):35–65.

Martins, P. and Correia, L. (2016). Fiscal forecasts and slippages: The role of the SGP and domestic fiscal frameworks. *Comparative Economic Studies*, 58(2):226–253.

Rogoff, K. (1990). Equilibrium political budget cycles. *American Economic Review*, 80(1):21–36.

Romer, C. D. and Romer, D. H. (2010). The macroeconomic effects of tax changes: Estimates based on a new measure of fiscal shocks. *American Economic Review*, 100(3):763–801.

Smith, R. (2007). An overview of the Tesseract OCR engine. In *Ninth International Conference on Document Analysis and Recognition (ICDAR 2007)*, volume 2, pages 629–633.

Spence, M. (1973). Job market signaling. *Quarterly Journal of Economics*, 87(3):355–374.

von Hagen, J. and Harden, I. J. (1995). Budget processes and commitment to fiscal discipline. *European Economic Review*, 39(3–4):771–779.

Appendices

Appendix A

Table .9: Summary statistics: fiscal variables

Variable	N	Mean	Std. dev.	Min	Max
Dependent variables: implementation error					
Headline balance	162	−0.59	2.81	−10.57	11.31
CAPB	162	−0.37	2.00	−6.94	10.68
Cyclically adjusted balance	162	−0.34	2.00	−7.07	10.32
Structural balance	162	−0.42	1.66	−6.54	3.46
Debt-to-GDP ratio (change)	162	0.42	6.64	−17.33	31.00
Revenue	162	0.30	1.10	−3.04	3.52
Expenditure	162	0.89	2.91	−12.05	13.97
Dependent variables: first-release change					
Headline balance	162	0.04	3.08	−10.87	10.00
CAPB	162	−0.12	2.05	−7.58	9.78
Cyclically adjusted balance	162	−0.02	2.03	−7.51	9.06
Structural balance	162	−0.09	1.76	−6.74	6.15
Debt-to-GDP ratio (change)	162	−0.36	6.78	−23.32	25.14
Revenue	162	0.11	1.00	−2.86	3.42
Expenditure	162	0.07	3.24	−10.09	12.84
Planned change (autumn $t-1$ vintage)					
Headline balance	162	0.63	1.41	−2.89	5.94
CAPB	162	0.25	1.13	−2.27	4.57
Cyclically adjusted balance	162	0.32	1.12	−2.19	4.62
Structural balance	162	0.33	1.15	−2.52	4.62
Debt-to-GDP ratio (change)	162	−0.78	2.65	−9.22	6.99
Revenue	162	−0.19	0.68	−3.56	2.46
Expenditure	162	−0.82	1.52	−7.12	3.54
Dependent variables: change in latest data					
Headline balance	179	0.13	2.78	−10.73	6.70
CAPB	179	−0.09	1.95	−6.66	6.68
Cyclically adjusted balance	179	0.01	1.92	−6.65	5.98
Structural balance	179	−0.03	1.86	−6.26	7.40
Debt-to-GDP ratio (change)	179	−0.92	6.73	−27.36	26.21
Revenue	179	0.02	1.11	−6.94	2.85
Expenditure	179	−0.11	2.91	−8.48	11.62

Notes: Statistics are computed on the observations used in the regressions. All variables are annual changes in percentage points of GDP. The implementation error is the first-release change (spring vintage of $t+1$) minus the planned change (autumn vintage of $t-1$).

Table .10: Summary statistics: controls and plan rhetoric

Variable	N	Mean	Std. dev.	Min	Max
Controls					
Debt-to-GDP ratio, level in $t-1$ (autumn $t-1$ vintage)	162	77.85	40.47	7.96	207.11
Output gap, t (autumn $t-1$ vintage)	162	−0.42	1.99	−7.47	4.03
Election year (0/1)	162	0.23	0.42	0.00	1.00
Plan rhetoric (within-country z-scores)					
Consolidation rhetoric (z)	162	0.15	1.01	−2.00	3.16
Credibility rhetoric (z)	162	0.02	0.95	−1.88	2.44
Revenue measures (z)	162	0.07	0.98	−1.81	3.06
Expenditure measures (z)	162	0.08	0.95	−1.93	2.43
Plan rhetoric (raw dictionary scores)					
Consolidation (per 1,000 words)	162	3.52	3.01	0.37	26.06
Credibility (per 1,000 words)	162	0.40	0.36	0.00	2.23
Revenue measures (per 1,000 words)	162	2.25	1.20	0.00	6.92
Expenditure measures (per 1,000 words)	162	1.46	0.81	0.28	4.59

Notes: Statistics are computed on the observations used in the regressions. The debt ratio is a level in percent of GDP and the output gap is in percent of potential GDP, both from the autumn vintage of year $t-1$. Rhetoric z-scores are standardised within country, so their mean is close to zero and their standard deviation close to one by construction; raw scores are dictionary matches per 1,000 words.

Appendix B

Table .11 lists the English dictionary, which is used to score the large majority of plans. Entries are phrases; a trailing * matches any ending, and matching is insensitive to case, accents, hyphens and apostrophes. Table .12 reports the phrases that account for most matches in the English-language plans. Dictionaries for the other seven languages are reported in the Online Appendix.

Table .11: English fiscal dictionary

Concept (entries)	Dictionary entries
Consolidation (42)	*fiscal consolidation*, *budgetary consolidation*, *fiscal discipline*, *budgetary discipline*, *fiscal prudence*, *fiscal responsibility*, *deficit reduction*, *deficit target**, *reducing the deficit*, *reduce the deficit*, *debt sustainability*, *fiscal sustainability*, *sustainability of public finances*, *sound public finances*, *debt to gdp*, *debt ratio*, *debt reduction*, *structural deficit*, *structural balance*, *primary surplus*, *primary balance*, *medium term objective*, *medium term budgetary objective*, *stability and growth pact*, *stability programme*, *convergence programme*, *excessive deficit*, *maastricht criteria*, *fiscal compact*, *balanced budget*, *budgetary balance*, *fiscal adjustment*, *structural adjustment*, *fiscal effort*, *structural effort*, *fiscal space*, *debt brake*, *fiscal rule**, *expenditure rule*, *net expenditure path*, *adjustment path*, *medium term fiscal structural plan*
Credibility (26)	*credibility*, *credible*, *track record*, *transparency*, *accountability*, *meet the target**, *meeting the target**, *compliance with the target**, *european commitments*, *honour our commitments*, *honouring our commitments*, *fiscal council*, *independent fiscal institution**, *court of auditors*, *prudent forecast**, *realistic forecast**, *prudent assumption**, *slippage**, *contingent liabilities*, *fiscal risk**, *budgetary risk**, *market confidence*, *confidence of the markets*, *performance budgeting*, *endorse** *the macroeconomic*, *deviation from the target**
Revenue-based measures (33)	*tax revenue**, *additional revenue**, *revenue raising*, *revenue measure**, *tax increase**, *tax rise**, *tax base*, *broadening the tax base*, *base broadening*, *tax compliance*, *tax evasion*, *tax fraud*, *tax avoidance*, *fight against fraud*, *tax collection*, *tax reform*, *tax expenditure**, *tax loophole**, *tax burden*, *vat increase*, *vat rate**, *social security contributions*, *social contributions*, *privatisation**, *privatization**, *asset sale**, *non tax revenue**, *excise dut**, *new tax*, *surtax*, *surcharge*, *extraordinary contribution*, *windfall tax*
Expenditure-based measures (34)	*expenditure restraint*, *spending restraint*, *spending cut**, *expenditure cut**, *expenditure control*, *spending control*, *expenditure reduction*, *reduction in expenditure*, *reducing public spending*, *austerity*, *savings measure**, *efficiency saving**, *budgetary savings*, *spending review**, *expenditure review**, *expenditure ceiling**, *spending ceiling**, *expenditure containment*, *cost containment*, *expenditure rationalisation*, *wage bill*, *compensation of employees*, *wage freeze*, *pay freeze*, *hiring freeze*, *recruitment freeze*, *staff reduction**, *pension reform*, *subsidy reform*, *interest expenditure*, *interest burden*, *intermediate consumption*, *captive appropriations*, *expenditure side measure**

Notes: Number of entries in parentheses. Very generic descriptors (e.g. "public debt", "net borrowing") are excluded because they appear throughout every budget document. No entry belongs to more than one concept.

Table .12: Most frequent dictionary matches in English-language plans

Consolidation			Credibility		
Phrase	Matches	%	Phrase	Matches	%
structural balance	4,901	18.5	transparency	403	22.5
stability programme	4,272	16.1	fiscal council	379	21.1
primary balance	2,102	7.9	contingent liabilities	339	18.9
stability and growth pact	1,955	7.4	credible	132	7.4
debt ratio	1,588	6.0	accountability	66	3.7
debt to gdp	1,327	5.0	slippages	65	3.6
excessive deficit	1,300	4.9	credibility	61	3.4
debt reduction	837	3.2	fiscal risks	59	3.3
medium term budgetary objective	808	3.1	independent fiscal institution	48	2.7
structural adjustment	793	3.0	track record	45	2.5
fiscal effort	666	2.5	court of auditors	43	2.4
adjustment path	592	2.2	meet the target	21	1.2
structural deficit	571	2.2	endorsed the macroeconomic	16	0.9
medium term objective	530	2.0	fiscal risk	13	0.7
fiscal sustainability	373	1.4	european commitments	12	0.7
All matches	26,464	100.0	All matches	1,793	100.0

Revenue-based measures			Expenditure-based measures		
Phrase	Matches	%	Phrase	Matches	%
revenue measures	1,935	17.1	interest expenditure	2,670	37.9
tax burden	1,591	14.1	compensation of employees	1,521	21.6
social contributions	1,322	11.7	intermediate consumption	1,272	18.1
tax revenue	1,041	9.2	pension reform	327	4.6
social security contributions	874	7.7	spending review	309	4.4
tax revenues	495	4.4	wage bill	166	2.4
tax reform	430	3.8	spending reviews	152	2.2
excise duty	371	3.3	expenditure review	77	1.1
tax base	364	3.2	expenditure ceilings	72	1.0
privatisation	346	3.1	expenditure ceiling	71	1.0
excise duties	283	2.5	expenditure cuts	54	0.8
vat rate	260	2.3	expenditure control	45	0.6
tax compliance	212	1.9	savings measures	36	0.5
tax evasion	204	1.8	expenditure restraint	35	0.5
tax collection	169	1.5	spending cuts	30	0.4
All matches	11,291	100.0	All matches	7,045	100.0

Notes: The 15 most frequent matched phrases per concept, across all English-language documents in the corpus (plans and Commission documents). "%" is the share of all matches of the concept. Inflected forms matched by the same wildcard entry are listed separately.